Improving understanding of farming approaches using Virtual Reality (VR) simulations.

Oluwakemi Akanle

Institution(s)

Advisor: Dr. Stefan Robila

Computational Sensing Laboratory, School of Computing, Montclair State University

Introduction

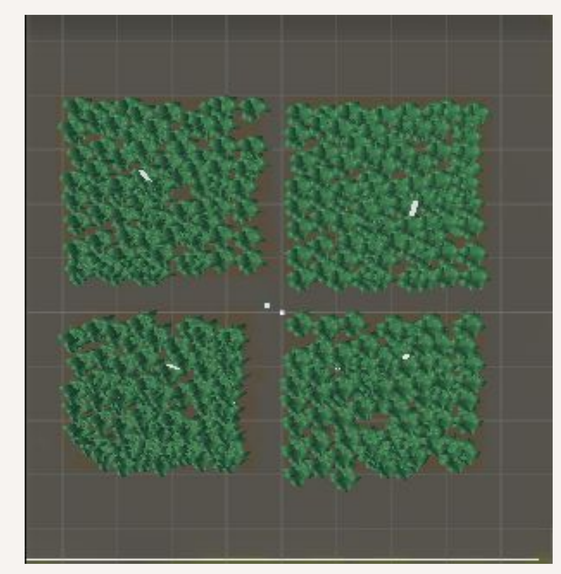
Farming contributes over 220 billion dollars to the Gross Domestic Product (GDP) and employs 2.6 million people. while fueling a food and beverage industry (that includes manufacturing, sales, and services) totaling 1,7 trillion $ and employing over 22 million people. Through this poster, we present the significance of farming as a fundamental aspect of human existence and emphasize the importance of ethical farming practices in the United States.

Objectives

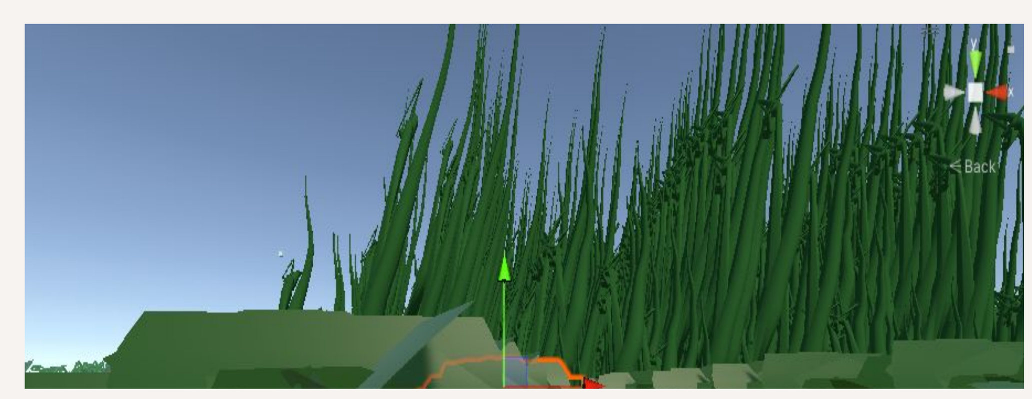
- Create a farming simulation that has that showcases / visualizes different farming types: R.U.N - Rural, Urban, and Native.
- Showcase the how each farming type can help the other using Unity and Virtual Reality (VR).

Results

- **Particle Systems**
  - The particle systems have 9 cubes over the following terrains they are timed differently to showcase water intake of the soil.
  - **Urban Farming**
    - Urban is 90% of the way done.
    - Vegetation includes: cabbage patch, cornfield, and radishes.
  - **Rural Farming**
    - Is 50% of the way there
    - Types of vegetation showcased : radishes and the cornfield.
  - **Native Farming**
    - Many of the "Native" lands like Arizona and Nevada are consistently having troubles with droughts
    - 20% of the way done
    - Vegetation showcased : wheat and wild corn.

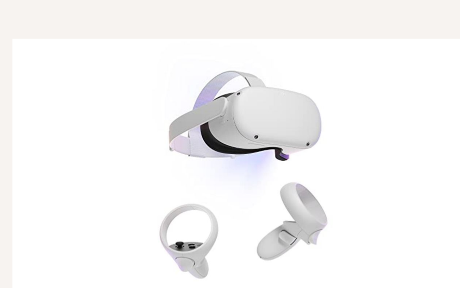





Materials and Methods

- **Computer Environment**
  - Cloud computing used for scene creation and hosting.
  Cloud based virtualization enables better resource utilization and flexibility in scaling resources up or down as needed.
  - Unity programming environment
  Virtualization creates Game Objects, Scenes, Physics Simulation, UI Elements, and Particle systems.
- **Virtual Reality (VR) Materials**
  - Oculus Headset
  - XPS Computer
  - Unity / 3D models / Assets

unity



- **Data**
  - Asset Importing for 3d models.
  - Unity Scene creation is used to create the environments , from camera view to lighting.
  - Unity usually uses C#
  - Resource Management reduces memory overhead, especially for resource-intensive projects such as games and simulations.

Conclusions / Future Directions

Simulation showcases the three different farming types in regards to R.U.N, Rural Urban, and Native farmland.

For future directions I will need to build upon Native farmland from to truly present how little water enter the land.

I would also like to create a weather machine in R.U.N that can will showcase natural disaster, predict accurate rain patterns in real time, and cloud movement.

I would also like a farming simulation of all 50 states.

References and/or Acknowledgements

I would like to acknowledge receiving a research stipend from the National Science Foundation through the Garden State-LSAMP.